Ratios, Proportions & Percentages

ClipSAT Printable Practice Suite (clipsat.org) • Topic 2.01

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 [Foundational] • Target: Simplifying ratios

Simplify the ratio $18 : 24$.

2 [Foundational] • Target: Simplifying ratios

Simplify the ratio $45 : 60 : 75$.

3 [Foundational] • Target: Simplifying ratios

Express "12 boys to 18 girls" as a ratio in simplest form.

4 [Foundational] • Target: Sharing a quantity in a given ratio

Share 150 in the ratio $2 : 3$.

5 [Foundational] • Target: Sharing a quantity in a given ratio

Share 220 among three people in the ratio $3 : 4 : 4$.

6 [Foundational] • Target: Sharing a quantity in a given ratio

A prize of 900 riyals is shared between Sara and Layla in the ratio $5 : 4$. Find each share.

7 [Foundational] • Target: Sharing a quantity in a given ratio

Divide 84 kg of flour in the ratio $1 : 2 : 4$.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Equivalent ratios and proportions

If $3 : 5 = x : 20$, find x.

9 [Intermediate] • Target: Equivalent ratios and proportions

If $a : 7 = 12 : 21$, find a.

10 [Intermediate] • Target: Equivalent ratios and proportions

A map has scale $1 : 50{,}000$. A distance of 3 cm on the map represents how many meters in reality?

11 [Intermediate] • Target: Equivalent ratios and proportions

Solve for y: $\frac{4}{9} = \frac{y}{63}$.

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12 [Intermediate] • Target: Percent of a number

Find 35% of 240.

13 [Intermediate] • Target: Percent of a number

What is 8% of 650?

14 [Intermediate] • Target: Converting between fractions, decimals and percentages

Write $\frac{3}{8}$ as a percent.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 [Challenging] • Target: Converting between fractions, decimals and percentages • Cross-Lesson: 3.01 Algebra Essentials

Write 0.45 as a fraction in lowest terms.

16 [Challenging] • Target: Converting between fractions, decimals and percentages • Cross-Lesson: 4.01 Geometry & Measurement

Write 62.5% as a fraction in lowest terms.

17 [Challenging] • Target: Converting between fractions, decimals and percentages • Cross-Lesson: 5.01 Quantitative Comparisons

Write $\frac{7}{40}$ as a decimal.

18 [Challenging] • Target: Percentage increase and decrease • Cross-Lesson: 6.01 Sequences & Patterns

Increase 80 by 25%.

19 [Challenging] • Target: Percentage increase and decrease • Cross-Lesson: 7.01 Data Analysis & Probability

Decrease 150 by 20%.

20 [Challenging] • Target: Percentage increase and decrease • Cross-Lesson: 8.01 Word Problems & Reasoning

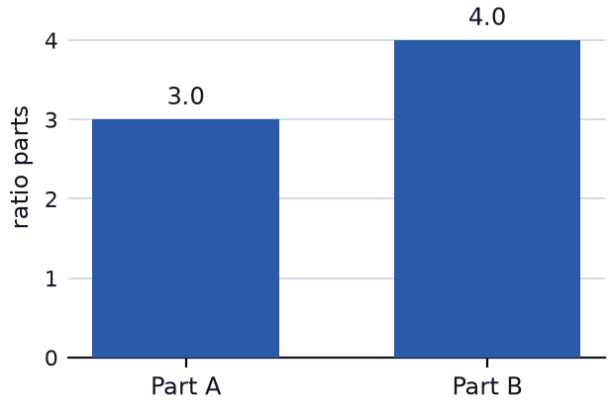
A salary of $4{,}000$ riyals increases by 6%. Find the new salary.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

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21 [Foundational] • Target: Sharing in a given ratio

The bar chart shows the ratio in which a total of 84 is to be shared. Find the larger share.



a. 36

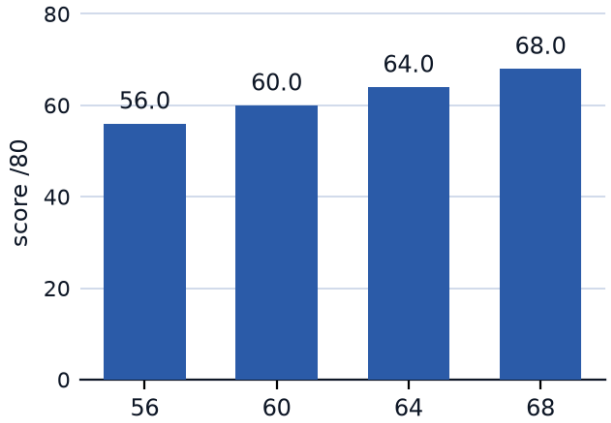
b. 21

c. 48

d. 63

22 [Foundational] • Target: Percent of a number

The bar chart shows four students' quiz scores out of 80. Which score is exactly 75%?



a. 68

b. 64

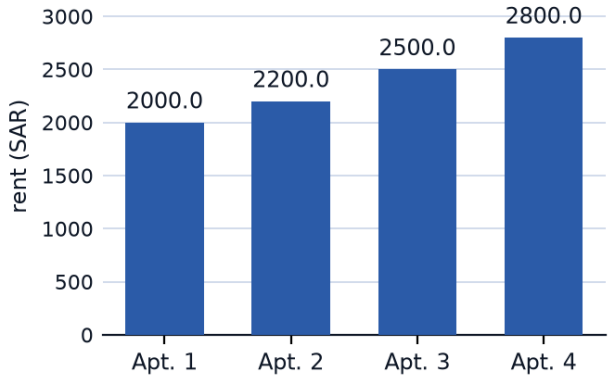
c. 60

d. 56

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23 [Foundational] • Target: Percent of a number

The bar chart shows monthly rent for four apartments. 8% of which apartment's rent equals 200 SAR?



a. Apt. 2

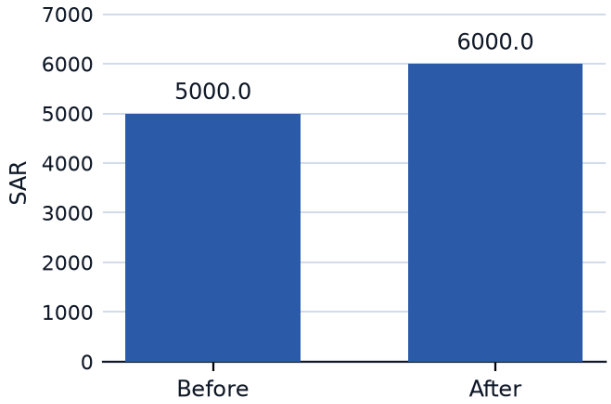
b. Apt. 1

c. Apt. 3

d. Apt. 4

24 [Foundational] • Target: Percentage increase & decrease

The bar chart shows a salary before and after a raise. What percentage increase does it represent?



a. 15%

b. 500%

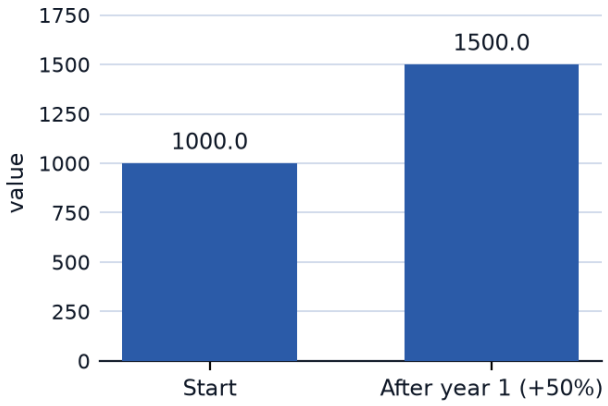
c. 20%

d. 25%

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25 [Foundational] • Target: Successive percentage changes

The bar chart shows a stock's value at the start and after rising 50% in year one. It then falls 50% in year two. Find its final value.



a. 500

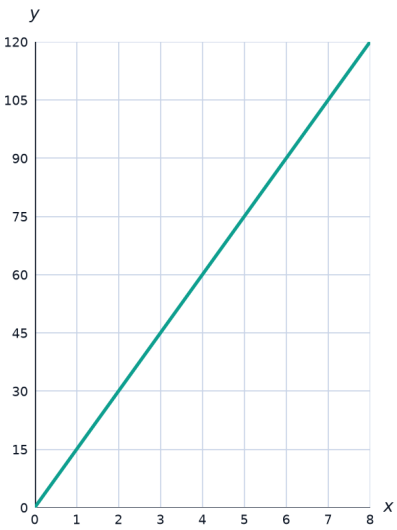
b. 900

c. 1,000

d. 750

26 [Foundational] • Target: Direct proportion

The graph shows the direct proportion between the number of hours x worked and pay y in SAR, $y = 15x$. How much is earned for 6 hours?



a. 105

b. 90

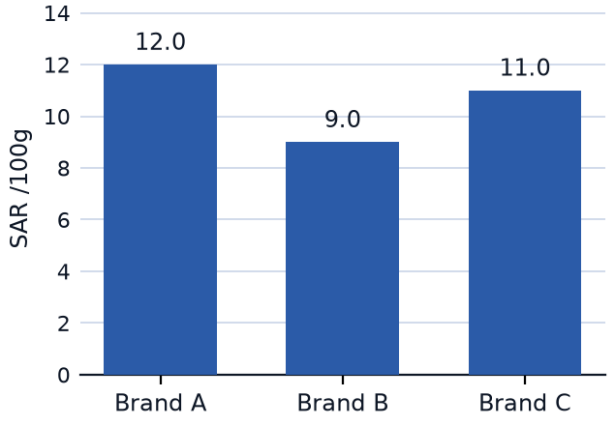
c. 75

d. 21

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27 [Foundational] • Target: Best value & unit price

The bar chart shows the price per 100 g for three brands of coffee. Which brand offers the best value?



a. Brand B

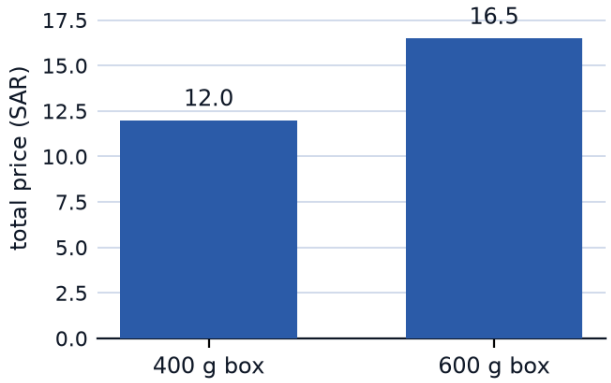
b. All are equal

c. Brand C

d. Brand A

28 [Foundational] • Target: Best value & unit price

The bar chart shows the total price of two cereal boxes of different sizes: 400 g and 600 g. Which has the lower price per 100 g?



a. Cannot be determined

b. They are equal

c. The 600 g box

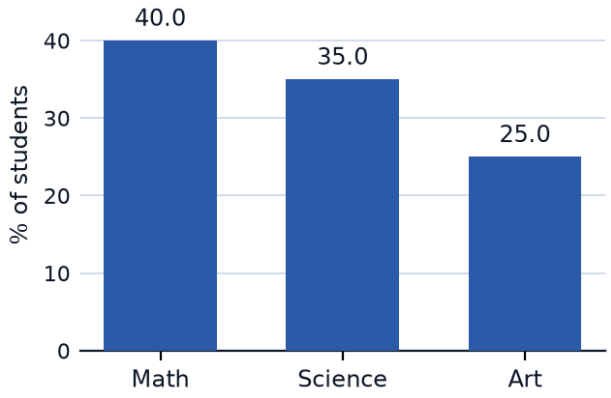
d. The 400 g box

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

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29 [Intermediate] • Target: Word problems

The bar chart shows subject preference percentages from a survey of 250 students. How many students prefer math?



a. 60

b. 90

c. 110

d. 100

30 [Intermediate] • Target: Simplifying & writing ratios

Simplify $18 : 45$.

a. $3 : 5$

b. $2 : 9$

c. $2 : 5$

d. $6 : 15$

31 [Intermediate] • Target: Simplifying & writing ratios

A recipe uses 250 g flour and 2 eggs. Write the ratio of flour (in grams) to eggs in simplest form.

a. $1 : 125$

b. $250 : 2$

c. $125 : 1$

d. $2 : 250$

32 [Intermediate] • Target: Simplifying & writing ratios

Express $0.75 : 2$ as a ratio of whole numbers in simplest form.

a. $3 : 4$

b. $3 : 8$

c. $75 : 2$

d. $75 : 200$

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33 [Intermediate] • Target: Sharing in a given ratio

Share 260 SAR in the ratio $2:3:5$. Find the smallest share.

a. 78 SAR

b. 52 SAR

c. 130 SAR

d. 65 SAR

34 [Intermediate] • Target: Sharing in a given ratio

Two business partners share a profit of $18{,}000$ SAR in the ratio $5:7$. Find the larger share.

a. $10{,}500$ SAR

b. $9{,}000$ SAR

c. $7{,}500$ SAR

d. $12{,}857$ SAR

35 [Intermediate] • Target: Sharing in a given ratio

A line of length 36 cm is divided in the ratio $4:5$. Find the length of the shorter piece.

a. 16

b. 4

c. 9

d. 20

36 [Intermediate] • Target: Sharing in a given ratio

A class of 40 students has boys and girls in the ratio $3:5$. How many boys are there?

a. 25

b. 8

c. 24

d. 15

37 [Intermediate] • Target: Equivalent ratios & scale

If $a:b=3:5$ and $a=18$, find b.

a. 30

b. 15

c. 10.8

d. 24

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

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38 [Challenging] • Target: Equivalent ratios & scale • Cross-Lesson: 6.01 Sequences & Patterns

A map has scale $1:50{,}000$. A distance of 3 cm on the map represents how many km in reality?

a. 15 km

b. 1.5 km

c. 150 km

d. 0.15 km

39 [Challenging] • Target: Equivalent ratios & scale • Cross-Lesson: 7.01 Data Analysis & Probability

Which ratio is equivalent to $4:9$?

a. $9:4$

b. $20:45$

c. $8:16$

d. $16:37$

40 [Challenging] • Target: Equivalent ratios & scale • Cross-Lesson: 8.01 Word Problems & Reasoning

In a photo, a model that is 1.8 m tall appears 6 cm high. Using the same scale, how tall (in m) is a building that appears 40 cm high in the photo?

a. 4.8

b. 24

c. 7.2

d. 12

41 [Challenging] • Target: Percent of a number • Cross-Lesson: 1.01 Arithmetic & Number Sense

150% of a number is 300. Find the number.

a. 250

b. 450

c. 200

d. 150

42 [Challenging] • Target: Fraction, decimal & percent conversion • Cross-Lesson: 3.01 Algebra Essentials

Write 0.06 as a percent.

a. 60%

b. 6%

c. 0.06%

d. 0.6%

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43 [Challenging] • Target: Fraction, decimal & percent conversion • Cross-Lesson: 4.01 Geometry & Measurement

Write 85% as a fraction in simplest form.

a. $\frac{17}{20}$

b. $\frac{8}{5}$

c. $\frac{17}{25}$

d. $\frac{85}{100}$

44 [Challenging] • Target: Fraction, decimal & percent conversion • Cross-Lesson: 5.01 Quantitative Comparisons

Which of these is largest: $\frac{3}{8}$, 0.4, or 37%?

a. 0.4

b. They are all equal

c. $\frac{3}{8}$

d. 37%

45 [Challenging] • Target: Percentage increase & decrease • Cross-Lesson: 6.01 Sequences & Patterns

A shirt costs 120 SAR after a 20% discount. Find the original price.

a. 150 SAR

b. 144 SAR

c. 100 SAR

d. 140 SAR

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